

ANIVIBE — Decision Support & Recommender Guide

Clean, Page-by-Page Architectural Breakdown (Zero Equations • Plain English)

Course: CSX/ITX 4207 Decision Support Systems **Institution:** Assumption University

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1. Home Page /

POPULARITY & PERSONALIZATION

User Goal: Immediate discovery. Delivers safe recommendations for new visitors and an evolving, personalized feed for users who rate shows.

1. Popular Right Now

POPULARITY-BASED RECOMMENDER

- **What it does:** Ranks anime by combining high rating scores with total review volume so titles with thousands of votes outrank one-off 5-star reviews.
- **Why we use it:** Solves the *Cold-Start problem*. When a new visitor has 0 ratings, the system provides verified, trusted community classics immediately.
- **User Benefit:** Instant viewing without having to search or set preferences.

2. Recommended For You

CONTENT-BASED FILTERING

- **What it does:** Learns your taste from your 1 to 5-star ratings. High ratings boost preferred genres; low ratings penalize unwanted themes.
- **Why we use it:** Automatically curates unrated anime that match your individual personality.
- **User Benefit:** Eliminates browsing fatigue by showing shows you are statistically likely to enjoy.

3. Interactive Onboarding Banner

DECISION SUPPORT PROMPT

- **What it does:** Prompts first-time visitors to rate their first 1–2 titles with quick-click stars.
- **Why we use it:** Actively guides users to calibrate the recommendation engine.

User Goal: Deciding whether a specific anime is worth your free time before committing hours or weeks to watch it.

1. More Like This (Netflix-Style Carousel)

ITEM-TO-ITEM CONTENT FILTERING

- **What it does:** Analyzes the genre profile of the anime on screen and recommends the closest sister titles in the catalog.
- **Why we use it:** Works independently of user rating history. If you loved this show, you can instantly find similar ones with one click.

2. Viewer Rating Distribution Histogram

COMMUNITY CONSENSUS SUPPORT

- **What it does:** Displays an amber bar breakdown showing the percentage of 5-star, 4-star, down to 1-star ratings.
- **Decision Support Value:** An average rating (e.g. 4.1) hides controversy. The histogram reveals whether an anime is universally loved or polarizing.

3. Browse & Find Page

Mode A: Catalog Discovery

EXPLORATORY DECISION SUPPORT

- **What it does:** Live search, genre filter pills, and multi-criteria sorting (by community popularity, review score, or title).
- **Why we use it:** Empowers self-guided browsing when users want to explore a single category.

Mode B: Find What Fits

KNOWLEDGE-BASED RECOMMENDER (KBR)

- **What it does:** Acts as an expert advisor. You set your situational boundaries:
 - **Episode Ceiling:** e.g. Maximum 26 episodes for a weekend binge.
 - **Quality Threshold:** e.g. Ratings $\geq$ 4.0 stars only.
 - **Genre Constraints:** Must include Mystery, strictly exclude Horror.
 - **Emotional Mood:** Exciting, Chill, Dark, or Emotional.
- **Explainable Verdicts:** Every card shows an itemized checklist of which rules passed.

4. Smart Match Page (Flagship)

/smart-match

1+1 HYBRID FUSION

User Goal: Solves the real-world dilemma: *"I want a show that fits my authentic personal taste, but also fits my available time and mood right now."*

1+1 Hybrid Fusion Recommender

HYBRID RECOMMENDER + DECISION SUPPORT

- **What it does:** Calculates two distinct scores and fuses them 50/50:
 - **50% Taste Score (Content-Based):** Does this match your past rating preferences?
 - **50% Fit Score (Knowledge-Based):** Does this fit your current episode limit and mood?
- **Why it's essential:** Pure taste filtering might recommend a 500-episode show when you only have one evening free. Pure constraint filtering might find a short show in a genre you dislike. The 1+1 Hybrid guarantees you get a title you love **and** have time to finish.
- **Dual-Score Transparency:** Result cards display separate **Taste Match %** and **Requirement Fit %** bars for complete clarity.

5. Decision Center

/decision-center

COMPARATIVE DECISION ANALYSIS

User Goal: Resolves final indecision when torn between 2 or 3 finalist anime.

Multi-Criteria Decision Matrix

MULTI-CRITERIA DECISION ANALYSIS (MCDA)

- **What it does:** Places up to 3 selected anime side-by-side in a comparative matrix.
- **Trade-Off Factors Compared:** Total episodes, hours to complete, community review counts, 5-star vs 1-star ratios, and genre overlap.
- **Decision Benefit:** Eliminates tab clutter and mental overload by presenting all trade-offs on one screen.

User Goal: Provides instructors and evaluators with empirical proof of model performance.

Empirical Benchmark Evaluation

SCIENTIFIC VALIDATION

- **Metrics Evaluated:** Precision (accuracy of matches), Recall (catalog relevance), Diversity (genre variety), and Coverage (% of catalog reachable).
- **Academic Finding:** Proves the **1+1 Hybrid** achieves superior precision while maintaining balanced diversity, validating the hybrid design.

Architecture At A Glance

Page	Recommender System	Decision Support Tool	Key Value
1. Home	Popularity + Personalized CBF	Cold-Start Decision Banner	Immediate start; personal feed as you rate
2. Anime Detail	Item-to-Item Similarity	Rating Histogram & Badges	Spot controversial vs loved titles
3. Browse & Find	Knowledge-Based Recommender	Rule Checklist & Filters	Match strict episode budget & mood
4. Smart Match	1+1 Hybrid (CBF + KBR)	Dual Taste vs Fit Match Badges	Picks what you love AND have time to watch
5. Decision Center	Comparative Candidate Ranking	Side-by-Side Trade-off Matrix	Resolve dilemmas between finalist shows
6. Benchmarks	Cross-Model Evaluation	Precision, Recall, Diversity Table	Objective proof for grading & defense

Academic Defense Summary for Evaluators

ANIVIBE demonstrates that a complete Decision Support and Recommendation System cannot rely on one paradigm alone. **Popularity** solves Cold-Start, **Content-Based Filtering** captures authentic taste, **Knowledge-Based rules** respect real-world time budgets, and the **1+1 Hybrid** unifies them into a transparent, optimal decision support platform.